

MegaFlash for Apple IIc and Apple IIc+ Istallation Guide

Thank you for purchasing the MegaFlash for a2s4000 add-on feature for the original Apple IIc. The following document will assist you installing the device in your machine.
The MegaFlash has been shipped to you preconfigured with the following:

- Storage
  - Volume 1 - A2Desktop 1.5
  - Volume 2 - Total-Replay 5.2
  - Volume 3 - 32MB empty ProDOS volume
  - Volume 4 - 32MB empty ProDOS volume
- RAMDISK enabled (400KB)
- AppleSoft FTP enabled
- Latest release of MegaFlash firmware

Included in the package:

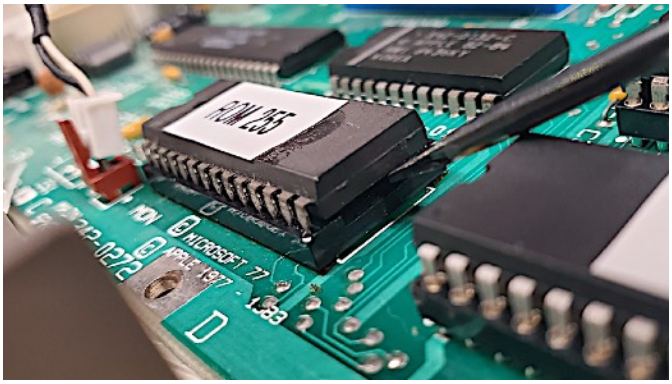
- MegaFlash board
- MegaFlash ROM chip for the Apple Iic

APPLE IIC A2S4000 AND A2S4100 (KILLIE KLIP) INSTALLATION INSTRUCTIONS

1. Determine current ROM version of your Apple IIc
   a. Power on your Apple IIc and type the following at the BASIC prompt:
      PRINT PEEK (64447)
   b. If the result is NOT 255, proceed to step 2
   c. If the result is 255, please refer to this procedure by BMOW to enable support for MegaFlash ROM
      a. https://www.bigmessowires.com/2015/05/29/apple-iic-rom-upgrade/
      b. YouTube video: https://youtu.be/NYHedg0-LMk
2. Disconnect your Apple IIc from power
3. Remove the cover from your Apple IIc. If unsure how, please search the web for videos appropriate for your model.
4. Once the top cover has been removed, lift away the keyboard. The pic below shows the board layout under the keyboard, with critical items labelled



5. With a small flat screwdriver, remove the MON ROM from the system. This is the chip labelled "ROM 255' in the previous picture. Be careful to insert the screwdriver between the chip and the socket below, into the small gap between the two. Carefully lever the chip out of the socket, if necessary also lifting from the other side.



6. Insert the MegaFlash ROM into the empty MON ROM socket. Take extra care to ensure the chip's notch is towards the speaker connector.
7. Pin the old ROM to the foam previously attached to the MegaFlash ROM. Place the ROM back in the anti-static bag and save.

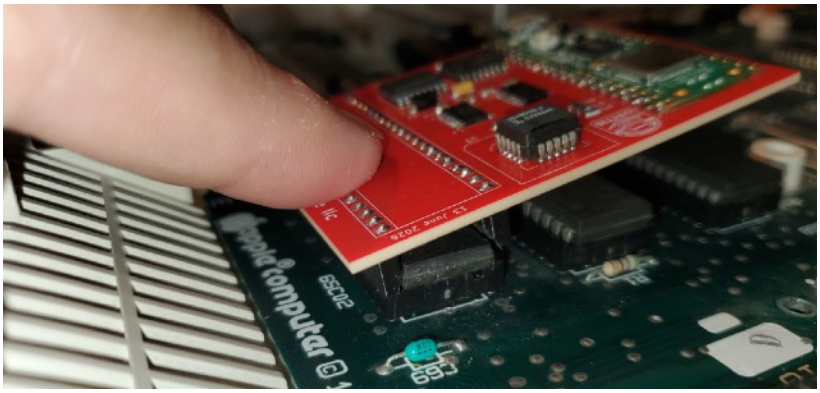
8. ATTACHING KILLIEKLIP STYLE MEGAFLASH

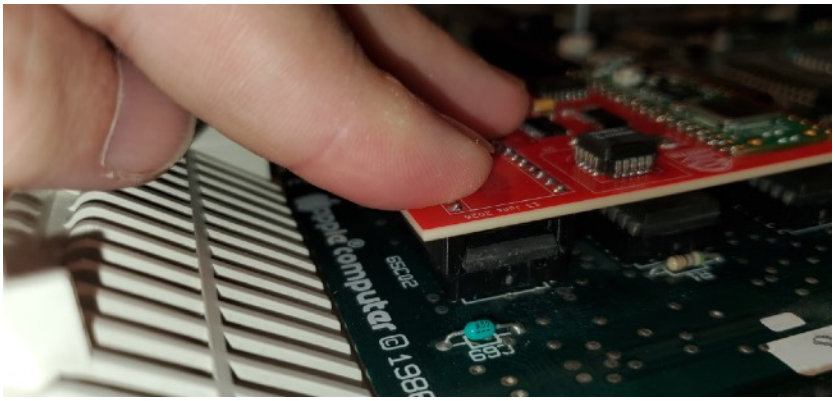
1. The The KillieKlip style MegaFlash mounts directly on top of the 6502 processor in the Apple IIc. It is no longer necessary to remove the processor from the motherboard. The plastic and metal pin feature on the underside of the MegaFlash board snaps directly onto the processor.



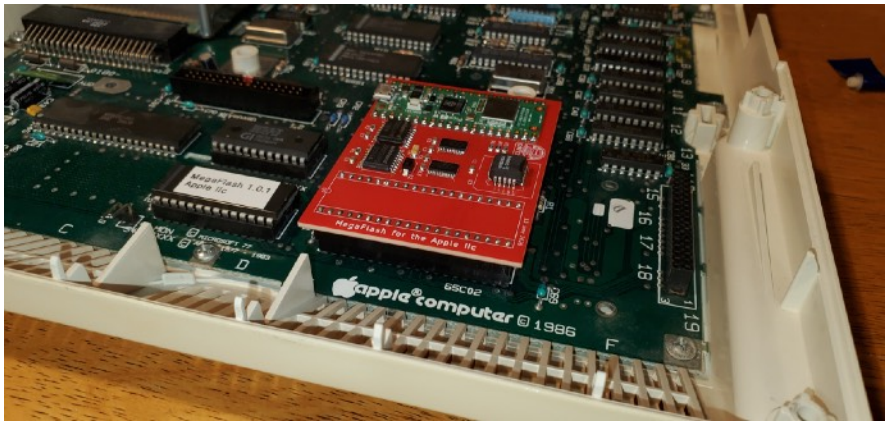


2. Identify the 6502 processor on your Apple IIc motherboard. For reference cross check in the picture on page 2, where all key components are clearly labelled. Turn the MegaFlash board so that the pico is towards the system floppy drive. First seat the front side of the board against the CPU as in the picture below.





3. Then gently press down on the board, right above the CPU to snap the board into place on the processor.
4. The board should snap firmly in place, and the cpu pins should no longer be visible.
5. Congratulations! You have installed MegaFlash to your Apple IIc



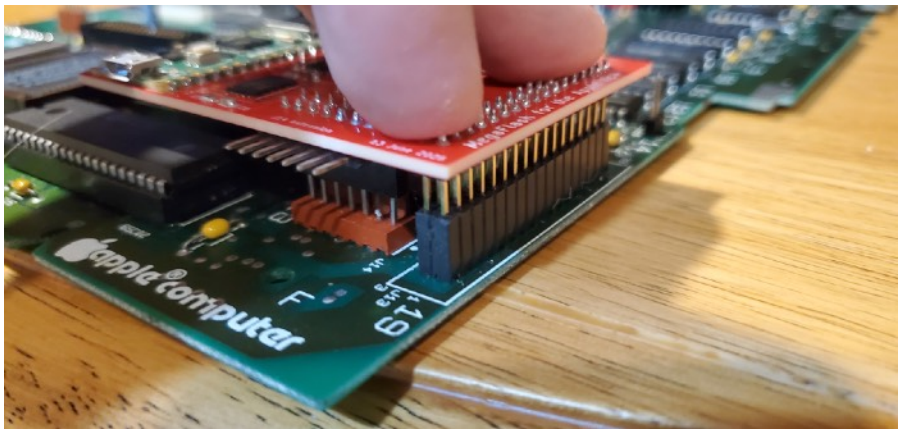
9. INSTALL MEGAFLASH TO THE APPLE IIC+

1. MegaFlash for the Apple IIc+ uses both the RAM expansion port on the right side of the motherboard, as well as the red/orange J14 connector perpendicular to it.
2. Note the 28 pin male pin connector on the underside of the board as well as the 7 pin female connector. These will mate with their counterparts on the motherboard.





3. Carefully align the board to directly connect to the motherboard.



4. Press down firmly until the board is fully seated.



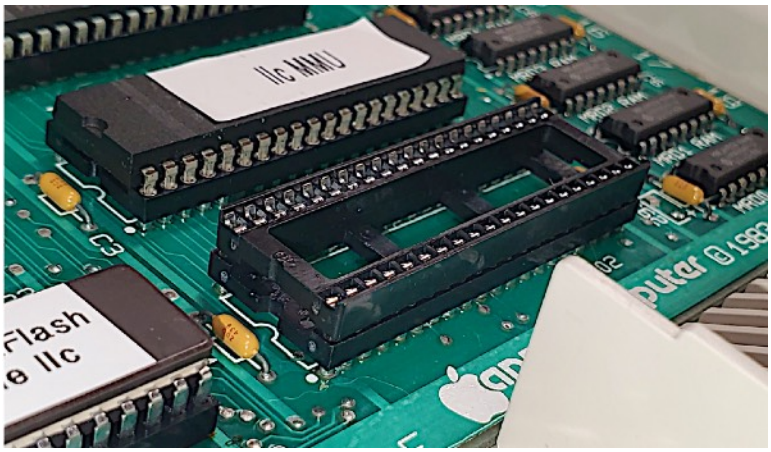
5. Congratulations! You have installed MegaFlash to your Apple IIc+!

10.INSTALLING THE ORIGINAL MEGAFLASH FOR THE A2S4000

1. Remove the 65c02 CPU from the motherboard. Similar to removing the ROM, carefully insert the flat screwdriver between the CPU and socket. Lift only a little, then repeat the same from the other side. Repeat raising from one then the other side until the chip has been removed. Care is needed to ensure the pins do not get bent.



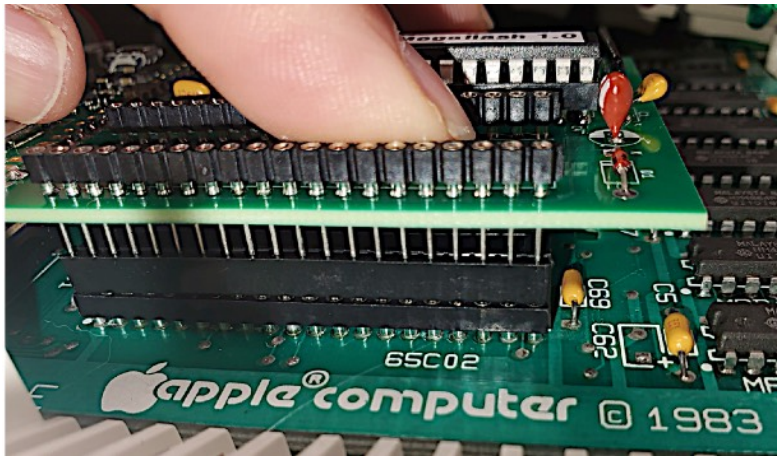
2. Insert the riser into the CPU socket as shown below



3. Insert the CPU into the empty socket on the MegaFlash with the protective anti-static foam still attached to the MegaFlash pins. Make sure the CPU notch is toward the Pico board, as in the picture below. It may take a bit of jiggling to get the CPU seated in the socket. Double check to ensure all pins are properly inserted into the holes, none of the pins are bent.

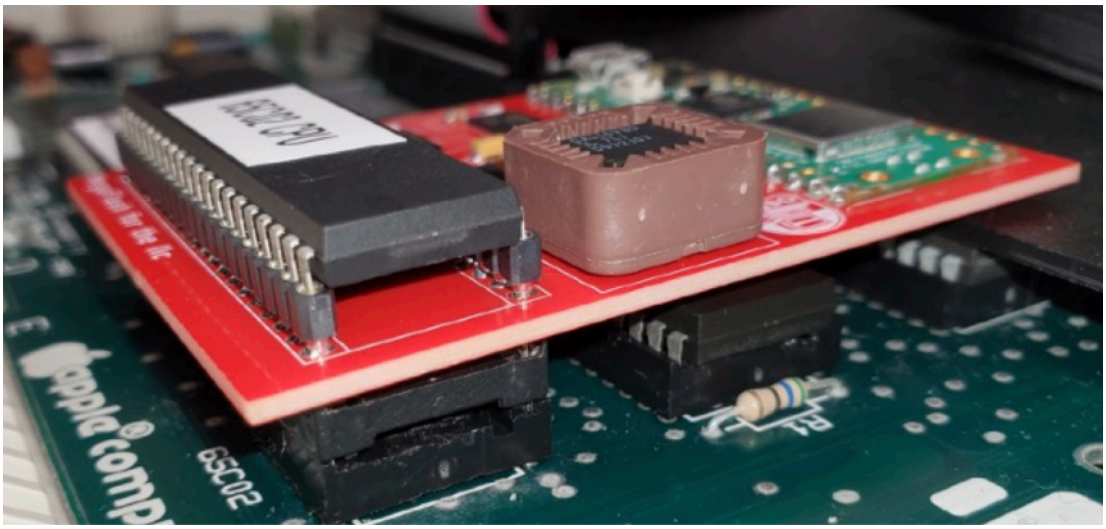


4. Before attaching the MegaFlash to the motherboard, remove the anti-static foam from the pins on the underside of the board. With the Pico on the left, carefully align the MegaFlash pins to the CPU riser socket. Press firmly on top of the CPU to seat it into the socket. Quite a bit of force may be needed.
5. Properly seated into the socket, only a mm or two of pin should be showing



6. Congratulations! The board has been installed! You may now close your Apple Iic





QUICK-START USER GUIDE FOR THE MEGAFLASH

With the MegaFlash installed, your Apple Iic will immediately boot into A2Desktop, a GUI file management environment for the Apple II line of computers. On the desktop you will see the A2Desktop 32MB volume, Total Replay, two empty volumes (Vol3 and Vol4), and a RAM disk. If you have a floppy in your drive or external drives attached, these will also show.
MegaFlash is fully compatible with all SmartPort devices attached to the external drive port. This includes FloppyEMU, FujiNet and others.

CONTROL PANEL QUICK START

To access the control panel, press and hold the Closed Apple key to the right of the keyboard space bar. Simultaneously press Ctrl-Reset and release. Keep holding down the Open Apple key until the boot selector panel appears on the monitor. Press 7 to enter the control panel.
By default all features will be enabled, except WiFi and clock.

1. With the down arrow, Scroll down to TimeZone and set for your location using the left/right arrows
2. Click twice more with the down arrow to select 'WiFi settings'. Press the Enter key
3. Enter your wifi network name and press enter
4. Enter your wifi password and press enter
5. Back in the control panel, click the down arrow one more time to select 'Test wifi/NTP'. Press Enter
6. If you correctly entered your WiFi credentials the test will show the IP assigned to the MegaFlash and confirm Network Time functionality. If the connection fails, return to the WiFi setup and re-enter your wifi name and credentials.
7. Once the WiFi connection is complete, scroll down to 'Save and Reboot'. This will reboot the machine and boot into A2Desktop. The top right corner of the screen will show the current time.
8. Basic setup is complete.

BOOTING FROM MEGAFLASH VOLUME 2

At the BASIC prompt type IN#4 to boot from volume 2 of MegaFlash

CONTROL PANEL DETAIL

DRIVES ENABLE

1. The flash ties and RAM disk can be individually enabled and disabled. Select 'Drives Enable' from the main menu to open the window. Press the number keys for the respective drives to enable or disable them. Enable drives will show a checkmark.
2. To save the changes, press Enter to return to the main menu.
   a. To save the settings, scroll down and select 'Save and Reboot'
   b. To discard the changes, press the ESC key.
3. Warning, the slot/drive mappings for the MegaFlash volumes will change after drives are enabled or disabled. Prefixes will stay the same, and will not change.

FORMAT

The MegaFlash drives and RAM disk can be formatted from the MegaFlash control panel. Two key features:

1. Volumes can be formatted to any arbitrary size up to 32MB

2. Formatting a volume will erase it. It may take up to 2 minutes to erase a flash drive.

MEGAFLASH HARDWARE/FIRMWARE INFORMATION

To view the hardware and firmware information, enter the MegaFlash boot menu by pressing and holding Closed-Apple while performing a Ctrl-Reset. Once the boot menu is displayed press Open-Apple-7 or Open-Apple-C. Keep holding the Open-Apple key until the device information screen appears.

ROM DISK

A ROM disk is provided for disaster recovery. The ROM disk is a Prodos volume and contains Copy II+, ADTPro and other utilities.

RAM DISK

The RAM disk can be enable from the Control Panel. The size of the RAM disk is 400 KB. The RAM disk can be used to transfer floppy disk images with ADTPro, transferring floppy disk images into the RAM disk. Once the data has been written to the RAM disk, it can be easily transferred to the persistent Flash storage volumes.

APPLESOFT BASIC FPU

Enabling the AppleSoft Basic FPU will enable the Apple IIc to offload math calculations to the Raspberry Pi Pico. This can improve complex math calculation performance by up to 8x.

REAL TIME CLOCK AND NETWORK TIME SYNC

The Raspberry Pi Pico has a real time clock function (RTC), but it does not support battery backup for persistent timekeeping. MegaFlash resolves this issue by performing NTP time requests over WiFi. If the Network Time Sync option is enabled, the system will automatically update the ProDOS time buffers, no drives are required.

To enable network time sync:

a. Configure the WIFI Settings from Control Panel
b. Run Test Wifi/NTP from Control Panel
c. If all tests are passed, enable the Network Time Sync option and setup the time zone.

If MegaFlash has been able to update the time via NTP, the current time will be displayed in the lower right corner in the Boot Menu and the Control Panel.

SLINKY EMULATION

With stock Apple IIc ROM3 or ROM4 firmware, the MegaFlash can emulate a 256KB Memory expansion car (aka Slinky).

DISK IMAGE TRANSFER

Disk images can be transferred to the MegaFlash in three ways:

1. ADTPro disk image transfers via Serial Port
2. MegaFlash terminal mode via USB
3. TFTP via WiFi

TERMINAL MODE DISK IMAGE TRANSFER

Disk images can be transferred to MegaFlash over USB. On MacOS, using the program like 'Screen', or on Windows machines using a program like 'TeraTerm'. A ProDOS ordered disk image (.po or .hdv) can be uploaded or downloaded using the XMODEM-CRC or XMODEM/1K protocol.

1) power off the Apple IIc
2) Connect a USB cable to the Pi Pico USB port
3) Connect a terminal session using your favorite terminal program to the Pico USB port
4) Press '1' to see details about your MegaFlash setup
5) Press '2' to select the volume to which to upload the disk image.
    a. Volume1 is the primary boot volume (PR#4)
    b. Volume2 is the secondary boot volume (IN#4)

DISK IMAGE TRANSFER VIA TFTP OVER WIFI

Dis images can be transferred between the MegaFlash and a network TFTP server over WiFi. Note that TFTP is not FTP. TFTP is a much simpler file transfer protocol. TFTP does not support directory listing and also does not require user authentication.

Note:

- The TFTP server and MegaFlash must be on the same subnet
- TFTP traffic cannot pass through firewall or NAT.
- Make sure blocksize option negotiation is enabled on the TFTPserver.
- When sending a file to the TFTP server, if the filename already exists on the TFTP server, it will be overwritten without any warning!

On Windows, Tfpd64 (https://pjo2.github.io/tftpd64/) server program is recommended. It has a GUI. You may need to run the app as administrator to save settings. Make sure Option negotiation is enabled.

On MacOS, consider using Pumpkin https://github.com/mterron/pumpkin

APPLE II DESKTOP

By default, Apple II Desktop versions before 1.5, copy itself to RAM Disk during start-up. This is not necessary for MegaFlash and should be disabled. This behavior can be disabled from Apple II Desktop Control Panel. Apple II Desktop shipped with your MegaFlash has this feature already disabled.

- Select Control Panel from Apple Menu.
- Open Options
- Select disable Copy to RAMCard.

UPDATING PI PICO BOARD FIRMWARE

MegaFlash by Ositis uses the Raspberry Pi Pico 2W exclusively. Your board was delivered with the latest firmware developed by Thomas Fok. Firmware updates from Thomas Fok is available here: https://github.com/ThomasFok/MegaFlash/releases

Alternate firmware is under development by Ositis. This MegaFlash fork is working on improving the networking stack and adding additional features. It is not as mature as Thomas' version. Use with caution! The current Ositis release has a completely rewritten network stack, and provides rudimentary UtherNet II emulation. This is still a work in progress. The currentrelease is available here: https://github.com/eositis/MegaFlash/releases

To upgrade the firmware:

1. Remove MegaFlash from your Apple IIc computer.
2. Press and hold the white button labeled BOOTSEL on the Pico board.
3. Connect the Pico board to a USB port of a computer while holding the button. Wait a few seconds, then release the white button.
4. A disk drive called RP2350 or RPI-RP2 should be mounted on your computer desktop.
5. Drag-and-drop the megaflash-pico2.uf2 firmware file downloaded from GitHub to the RP2350 drive. Wait until the file copy finishes.
6. Reinstall MegaFlash back to your Apple IIc computer carefully.

